Deposit Competition and Financial Fragility: Evidence from the US Banking Sector

Mark Egan, Ali Hortaçsu and Gregor Matvos

Harvard University, University of Chicago, Northwestern University, NBER

Bank Runs

- The recent financial crisis brought renewed attention to
  - The stability of the banking sector
  - Optimal bank regulation.
- Extensive theoretical literature
  - Banks *might* be unstable
  - Runs *might* be possible
  - Are forces strong enough to cause instability?
- Empirical Literature:
  - Detailed case studies of bank runs
    - Calomiris and Mason (2003), Soledad et al. (2001), Iyer and Puri (2012), Iyer and Ryan (2013)

Uninsured deposits and U.S. Commercial Banks

- Deposits are 75% of bank funding of U.S. commercial banks
  - Half are uninsured
- If uninsured depositors worry about bank default
  - Withdraw funds
  - Cause decline in bank profits
  - Increase chance of default
  - More withdrawals
- Are these forces strong enough to cause “runs”
  - Do uninsured depositors care about default?
  - Do they care enough to cause runs?
  - Consequences?

Quantitative model of bank runs

Model of the banking sector

- Bank run ingredients:
  - Run prone uninsured depositors
  - Endogenous bankruptcy decision
- IO ingredients:
  - Differentiated banks
  - Demand system
  - Oligopolistic competition b/w banks
- Apply to data on U.S. commercial banks

Dynamic: Leland meets Matutes - Vives

- Within period timing:
  1. Bank set interest rates for insured and uninsured deposits
  2. Consumers select where to deposit funds based on:
     - Interest rates
     - Probability of default
     - Bank services
  3. Banks experience a stochastic income/return shocks
  4. Banks decide whether to
     1. Repay debt & deposits
     2. or default

Demand

Depositors choose among differentiated banks:

M^N uninsured deposits:

$$u^N_{j,k} = \underbrace{\alpha^N i^N_{k,t}}_{interest} - \underbrace{\gamma \rho_{k,t}}_{default} + \underbrace{\delta^N_k + \varepsilon^N_{j,k}}_{banking\ services}$$

M^I insured deposits:

$$u^I_{j,k} = \underbrace{\alpha^I i_{k,t}}_{interest} + \underbrace{\delta^I_k + \varepsilon^I_{j,k}}_{banking\ services}$$

Banks compete for deposits by setting interest rates:

- $s^I_{k,t}$, $s^N_{k,t}$ market share of insured/uninsured deposits

Invest deposits and earn

- R_k on deposits
- $R_k \sim N(\mu_k, \sigma_k)$

Gross per period bank profits

$$M^I s^I_{k,t}(1 + R_{k,t} - c_k) + M^N s^N_{k,t}(1 + R_{k,t})$$

Every period have to repay:

- Deposits + interest: $M^I s_{k,t}^I \left(1 + i_{k,t}^I\right) + M^N s_{k,t}^N \left(1 + i_{k,t}^N\right)$
- Long term debt coupon b_k

Equity shortfall:

$$b_k - \left(\underbrace{M^I s_{k,t}^I}_{insured\, dep.} \underbrace{\left(R_{k,t} - c_k - i_{k,t}^I\right)}_{net\, int.\, margin} + \underbrace{M^N s_{k,t}^N}_{uninsured\, dep.} \left(R - i_{k,t}^N\right) \right)$$

Equity holders can choose to pay shortfall or default.

The equilibrium is characterized by three sets of optimality conditions

- Banks maximize value by setting interest rates
- Consumers maximize utility by choosing deposits
- Banks optimally elect to declare bankruptcy or continue operating

Beliefs are correct

Assume stationary setting

- Bank return shocks are i.i.d.
- Market parameters are constant
- Bankruptcy: bank retains same capital structure

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$\varepsilon^N_{j,k}$ and $\varepsilon^I_{j,k}$ i.i.d. T1EV

Market share of bank k:

$$s^I_{k,t}\left(i^I_{k,t}, \mathbf{i^I_{-k,t}}\right) = \frac{exp(\alpha^I i^I_{k,t} + \delta^I_k)}{\sum_{l=1}^K exp(\alpha i^I_{l,t} + \delta^I_l)},$$

$$s^N_{k,t}\left(i^N_{k,t}, \mathbf{i^N_{-k,t}}, \rho_{k,t}, \rho_{-k,t}\right) = \frac{exp(\alpha^N i^N_{k,t} - \rho_{k,t}\gamma + \delta^N_k)}{\sum_{l=1}^K exp(\alpha^N i^N_{l,t} - \rho_{l,t}\gamma + \delta^N_l)}.$$

Banks k defaults if:

$$b_k - \left(\underbrace{M^I s^I_{k,t} \left(R_{k,t} - c_k - i^I_{k,t} \right)}_{\pi \, insured \, dep} + \underbrace{M^N s^N_{k,t} \left(R_{k,t} - i^N_{k,t} \right)}_{\pi \, uninsured \, dep} \right) > \frac{1}{1+r} E_k$$

Firms default if $R_{k,t}$ falls below some $\bar{R}_k$

Default threshold $\bar{R}_k$ determined by:

$$\underbrace{b_k - \left(M^I s^I_{k,t} \left(\bar{R}_k - c_k - i^I_{k,t} \right) + M^N s^N_{k,t} \left(\bar{R}_k - i^N_{k,t} \right) \right)}_{\text{shortfall at threshold}}$$

$$\frac{1}{1+r} \underbrace{\left(M^I s^I_{k,t} + M^N s^N_{k,t} \right)}_{\text{total deposits}} \underbrace{\left(1 - \Phi \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right) \right)}_{\text{survival prob.}} \times$$

$$= \underbrace{\left(\left(\mu_k - \bar{R}_k \right) + \sigma_k \underbrace{\lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right)}_{\text{limited liability}} \right)}_{\text{expected return on deposits}}$$

- Interest rates maximize period profits
  - Envelope w/r to optimal default
- For insured

$$\underbrace{\underbrace{\mu_k}_{mean\,return} + \underbrace{\sigma_k \lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right)}_{limited\,liabillity}}_{mb} - \underbrace{\left(c_k + i^I_{k,t} \right)}_{mc} = \underbrace{\frac{1}{\left(1 - s^I \left(i^I_{k,t}, \mathbf{i}^{\mathbf{I}}_{-\mathbf{k},\mathbf{t}} \right) \alpha^I\right.}}_{mark-up}$$

- For uninsured:

$$\mu_k + \sigma_k \lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right) - i^N_{k,t} = \frac{1}{\left(1 - s^N \left(i^I_{k,t}, i^I_{-k,t}, \rho_{k,t}, \rho_{-k,t} \right) \alpha^N\right.}$$

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Bank optimality
Setting interest rates

- Interest rates maximize period profits
  - Envelope w/r to optimal default
- For insured

$$\underbrace{\underbrace{\mu_k}_{\textit{mean return}} + \underbrace{\sigma_k \lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right)}_{\textit{limited liabillity}}}_{mb} - \underbrace{\left(c_k + i^I_{k,t} \right)}_{mc} = \underbrace{\frac{1}{\left(1 - s^I \left(i^I_{k,t}, \mathbf{i^I_{-k,t}} \right) \alpha^I\right.}}_{mark-up}$$

- For uninsured:

$$\mu_k + \sigma_k \lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right) - i^N_{k,t} = \frac{1}{\left(1 - s^N \left(i^I_{k,t}, i^I_{-k,t}, \rho_{k,t}, \rho_{-k,t} \right) \alpha^N\right.}$$

Extensions

- Capital requirements
  - Risky asset
  - Safe asset
- Too big to fail
- Costly equity injections
- Direct bankruptcy cost

Estimation / Calibration

Model lends itself to empirical estimation/calibration

1. Estimate a deposit demand system
   - Use only consumer optimization
   - Recover the utility parameters α^I ,α^N and γ
2. Calibrate bank parameters
   - Use bank optimization
   - Recover profit parameters μ_k, σ_k, and c_k

Data

- Sample: 16/20 largest commercial U.S. banks, 2002-2013.
- Credit Default Swap (CDS):
  - Direct and daily market measure of the financial solvency
  - We use five year CDS spread data from Markit
- Deposit Level / Balance Sheet:
  - Insured and uninsured deposit levels from Call Reports
- Deposit Rate Data:
  - Certificate of deposit (CD) rates provided by RateWatch.

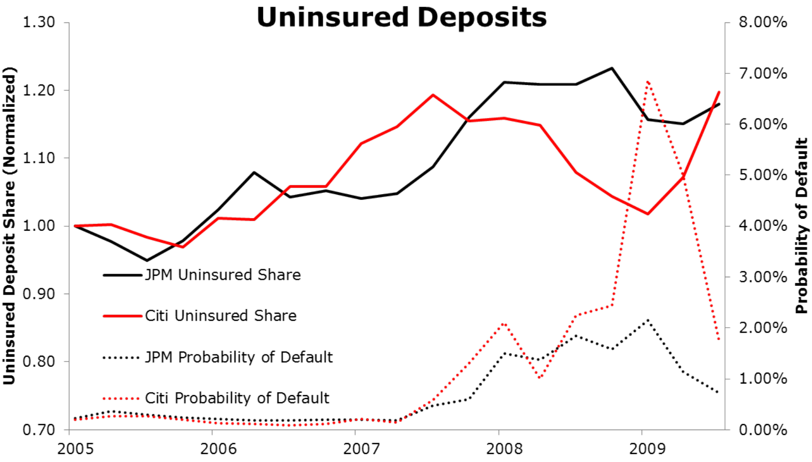
Three sources of demand for uninsured deposits:

- Interest rates
- Banking services
- Default probability

Problem: variation in unobservable quality of banking services

- Quality of services $\downarrow \Rightarrow$ Demand for deposits $\downarrow \Rightarrow$ Profits $\downarrow \Rightarrow$ Default prob. $\uparrow$

Deposit Rates vs Financial Distress: JPM vs Citi Bank



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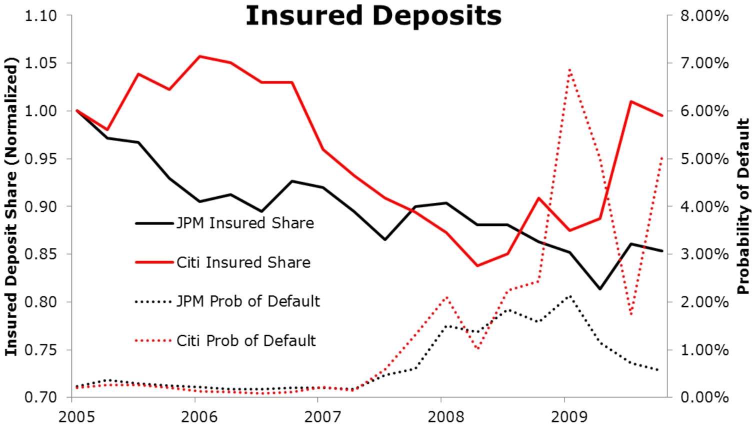


Financial distress and demand for deposits

Reduced form

	Diff. Deposits	Unins Deposits	Ins. Deposits
Prob of Default	-1.98**	-2.13*	-0.16
	(0.96)	(1.14)	(1.04)
Share Diff	X		
Quarter F.E	X	X	X
Bank F.E.	X	X	X
Observations	566	566	566
R-squared	0.949	0.970	0.948

Estimate demand system

- IV to deal with *changes* in unobserved quality
    - IV for default risk
        - Net amount of charged-off loans by a bank
    - IV for interest rates
        - Cost shocks (treasury rates)
        - Bank specific pass-through
- Placebo
    - Run IV for default probability of insured deposits

Financial distress and demand for deposits

IV

	Unins. Deposits		Ins. Deposits
Deposit Rate	18.21***	20.76***	58.79***
	(6.08)	(7.01)	(8.16)
Prob of Default	-10.66*	-11.27*	
	(5.83)	(6.49)	
Quarter F.E.	X	X	X
Bank F.E.	X	X	X
IV-0 (Pass-Through)	X	X	X
IV-1 (Loans)	X		
IV-2 (CMOs)		X	
Observations	564	564	566
R-squared	0.966	0.965	0.917

Financial distress and demand for deposits

IV placebo

	(1)	(2)
Deposit Rate	54.96***	58.76***
	(8.14)	(8.36)
Prob of Default	-2.86	3.16
	(5.33)	(5.30)
Quarter F.E.	X	X
Bank F.E.	X	X
IV-0 (Pass-Through)	X	X
IV-1 (Loans)	X	
IV-2 (CMOs)		X
Observations	566	566
R-squared	0.921	0.917

Demand Estimates

- Interest rate sensitivity for insured deposits: $\alpha^I = 58.79$
  - Elasticity of 0.64
- Interest rate sensitivity for uninsured deposits: α^I=18.33
  - Elasticity of 0.22
- Default risk sensitivity for uninsured deposits: $-\gamma$=-10.64
  - Elasticity of -0.36

Demand Estimates

- FE effects correlated with
  - ATM density
  - CFPB complaints (negative)
  - Insured vs. uninsured

Calibration

Calibrate bank parameters

- Recover profit parameters μ_k, σ_k, and c_k for each bank

- Use bank optimization (for each bank)
  - Interest rates on insured deposits
  - Interest rates on uninsured deposits
  - Default boundary

- Data
  - Demand estimates
  - Market shares
  - Deposit rates
  - Balance sheet

Non-interest cost of insured deposits

- Recover c_k for each bank

$$c_k = \left(i^N_{k,t} + \frac{1}{(1 - s^N_{k,t})\alpha^N} \right) - \left(i^I_{k,t} + \frac{1}{(1 - s^I_{k,t})\alpha^I} \right)$$

- Relative mark-up on insured and uninsured deposits

Volatility of returns

- Recover σ_k for each bank
- Threshold characterized by:

$$\textit{Shortfall at Threshold} = \textit{Continuation Value}$$

- Observe bankruptcy probability

$$\rho_k = \Phi\left(\frac{\bar{R}_k - \mu_k}{\sigma_k}\right)$$

- Continuation value:

$$\sigma_k \frac{\left(M^I s^I_{k,t} + M^N s^N_{k,t}\right)}{1+r} (1-\rho_k)\left(-\Phi^{-1}(\rho_k) + \lambda\left(\Phi^{-1}(\rho_k)\right)\right)$$

- Shortfall at default threshold

$$b_k - M^I s^I_{k,t}\left(\bar{R}_k - c_k - i^I_{k,t}\right) - M^N s^N_{k,t}\left(\bar{R}_k - i^N_{l,t}\right)$$

- Interest margin can be expressed as:

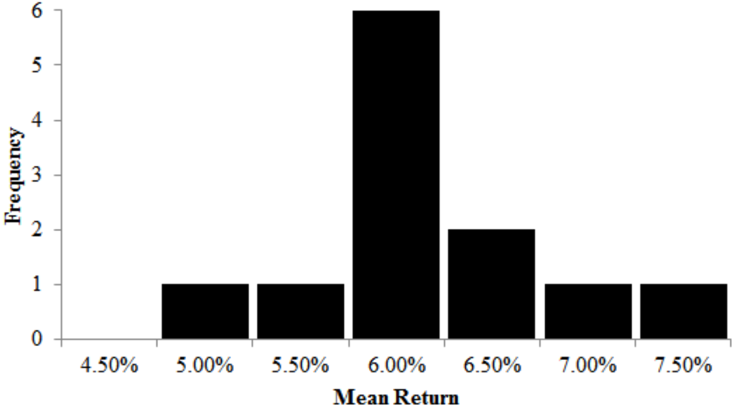
$$\left(\bar{R}_k - c_k - i^I_{k,t}\right) = \frac{1}{(1-s^I)\,\alpha^I} + \sigma_k\left(\Phi^{-1}(\rho_k) - \lambda\left(\Phi^{-1}(\rho_k)\right)\right)$$

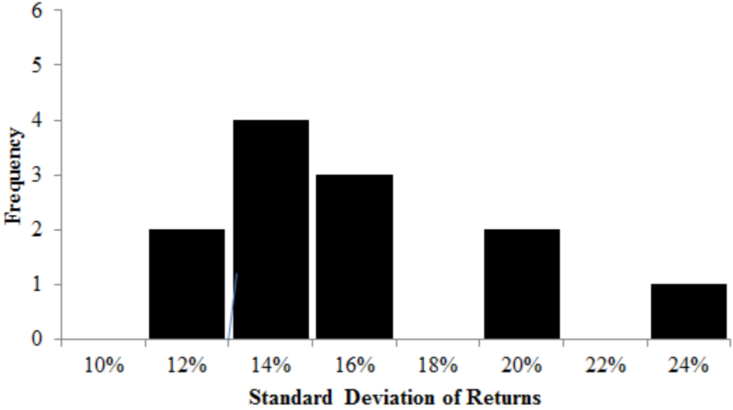
- Recover μ_k for each bank

$$\mu_k = i_{k,t}^N - \sigma_k \lambda \left(\Phi^{-1}(\rho_k) \right) + \frac{1}{(1 - s^N)\alpha^N}$$

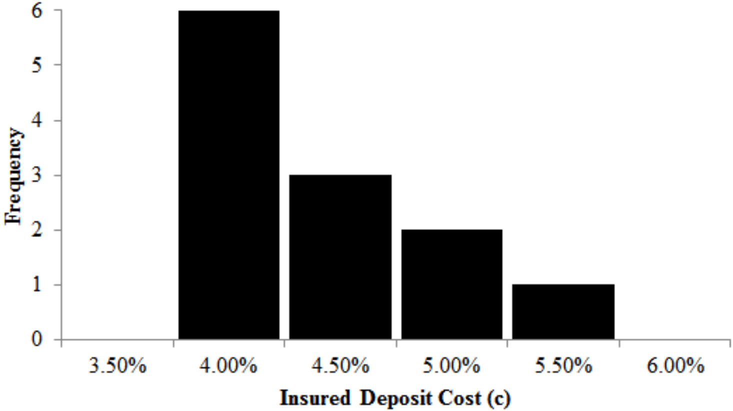
- Pass through of profits to consumers

Results: Mean Return




6
5
4
3
2
1
0
Frequency
10%
12%
14%
16%
18%
20%
22%
24%
Standard Deviation of Returns

Non-interest Costs



Is the Calibration Reasonable?

Non-interest costs 4%-5.5%

- For \$1,500 $\Rightarrow$ \$5-\$7 per month
- Bank fees for small account \$5-\$10

Interest margin

- Return on deposits - interest costs - non-interest costs
    - 2.3%-3.75%
- Hanson et al (2014), approximately 2%

Multiple equilibria

Potential for multiple equilibria.

- Uninsured deposits run-prone
- Bank solvency depends on deposits

Are estimated forces strong enough that multiple equilibria a possibility?

- We search for other equilibria in the model
  - Fix estimated technology, preferences
  - Change
    - Consumer choices (market shares)
    - Default probabilities
    - Interest rates
- Focus our attention to the five largest banks
- Analysis generalizes to the full set of banks

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Wachovia, March 2008

Realized equilibrium:

- Risk neutral probability of default, 3.3%

Alternative equilibrium:

- Risk neutral probability of default, 56%

Difference: consumer beliefs

We can find better equilibria (March, 2008):

Bank	Observed Eq.	Better Eq.
JPMorgan Chase	1.50%	0.28%
Citi Bank	2.11%	1.94%
Bank of America	1.82%	0.04%
Wells Fargo	1.50%	1.36%
Wachovia	3.28%	3.18%

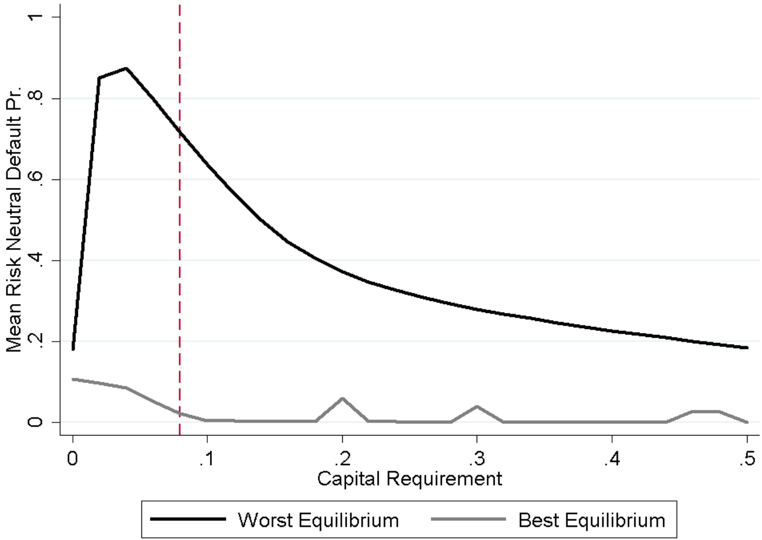
But not much better, and not many.

March, 2008:

		Other Equilibria				
	Obs.	(2)	(3)	(4)	(5)	(6)
JPMorgan Chase	1.50%	50.65%	4.39%	3.39%	2.80%	2.80%
Citi Bank	2.11%	4.74%	51.74%	3.85%	3.28%	3.29%
Bank of America	1.82%	2.58%	2.54%	56.55%	0.99%	1.33%
Wells Fargo	1.50%	4.73%	4.91%	3.44%	47.89%	3.27%
Wachovia	3.28%	4.99%	5.00%	4.29%	3.62%	55.99%

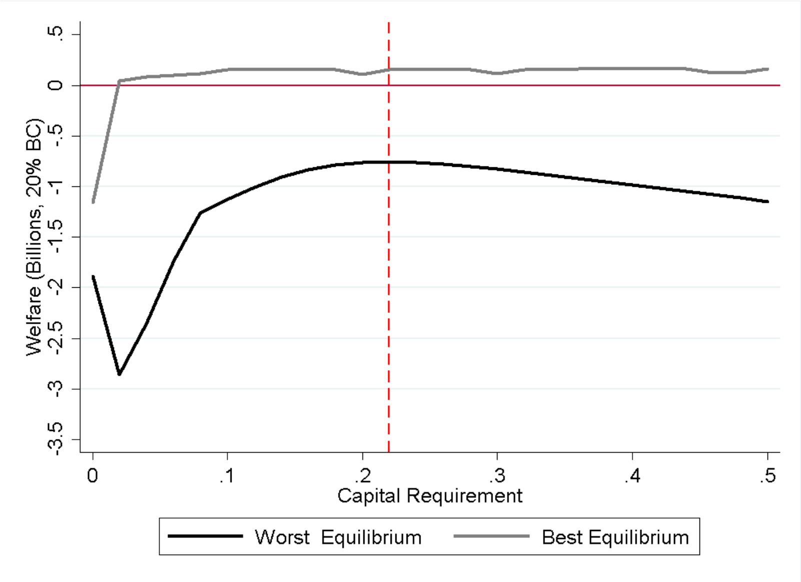
Counterfactuals

- Use the model to compute many policy counterfactuals
  - Increased capital requirements
  - Increased deposit insurance
  - Interest rate limits
  - Risk limits
- Compute:
  - Bank stability
  - Welfare:
    - Consumer surplus
    - Equity value (annualized)
    - FDIC expected costs



Mean Risk Neutral Default Pr.
0
.2
.4
.6
.8
1
0
.1
.2
.3
.4
.5
Capital Requirement
Worst Equilibrium
Best Equilibrium

Capital requirements
Welfare



Optimal capital requirement

- Multiple equilibria / capital requirement
- Difficult to say
  - Which equilibrium will be played?
  - Distribution over equilibria?
- Uncertainty averse planner (Gilboa and Schmeidler 1989)
  - No restrictions on distribution of priors over equilibrium realizations
  - Maximizing the worse equilibrium outcome
    - Capital requirement 31%

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  - Maximizing the worse equilibrium outcome
    - Capital requirement 31%

Conclusion

The demand estimates suggest that

- Uninsured depositors are sensitive to financial distress
- Insured depositors are not

Interdependence of deposit demand and solvency creates instability

- "Bad" equilibria exist for each bank
- Banks were at a relatively "good" equilibrium during the crisis

Competition for deposits leads to spillovers

- Even without interbank credit exposure!

Bank Regulations:

- Capital requirements
- Limiting rates
- Increased deposit insurance
- Bank risk limits